

TURKU ARTUKAINEN, Finland

WMO: 027730

Lat: 60.4544N

Lon: 22.1786E

Elev: 30

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.8	3.6	-6.2	4.0	-0.8	-0.4	5.4	4.6	20.8	26.9	18.5	28.3	4.4	30	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	81.6	65.8	78.0	64.7	74.7	63.0	69.1	77.4	66.9	74.8	64.7	72.3	7.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.3	96.9	72.9	64.0	89.4	70.4	61.8	82.5	68.0	33.2	77.2	31.4	74.4	29.8	72.2	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.2	14.3	12.9	-5.1	85.4	7.6	4.6	-10.6	88.7	-15.1	91.4	-19.4	93.9	-25.0	97.2		
			-5.6	70.1	7.5	2.6	-11.0	72.0	-15.4	73.5	-19.7	75.0	-25.2	76.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	43.7	23.5	26.0	30.9	40.5	51.9	58.7	64.9	62.2	54.2	42.9	37.2	30.6	(6)
(7)		DBStd	15.66	10.20	9.80	7.20	5.66	7.64	5.85	5.48	4.93	5.55	6.35	7.18	9.03	(7)
(8)		HDD50	3686	821	672	593	290	65	4	0	0	25	228	385	603	(8)
(9)		HDD65	7882	1286	1092	1058	735	410	205	69	114	326	684	835	1068	(9)
(10)		CDD50	1396	0	0	0	5	126	265	463	378	150	9	0	0	(10)
(11)		CDD65	117	0	0	0	0	5	16	67	28	1	0	0	0	(11)
(12)		CDH74	881	0	0	0	0	64	147	528	141	1	0	0	0	(12)
(13)		CDH80	179	0	0	0	0	5	21	124	29	0	0	0	0	(13)
(14)	Wind	WSAvg	6.4	6.0	6.4	6.6	6.2	6.3	6.7	6.2	6.3	5.9	6.4	6.8	6.7	(14)
(15)	Precipitation	PrecAvg	28.5	2.4	1.7	1.7	1.5	1.4	2.0	2.9	3.2	2.7	2.8	3.1	2.8	(15)
(16)		PrecMax	38.0	4.3	4.2	4.1	4.0	3.5	4.1	5.7	7.5	6.4	6.8	6.4	7.2	(16)
(17)		PrecMin	20.9	0.4	0.1	0.2	0.1	0.2	0.3	0.0	0.0	0.5	0.8	0.5	0.3	(17)
(18)		PrecStd	3.8	1.1	1.0	0.9	0.9	0.9	1.1	1.6	1.7	1.4	1.3	1.5	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	43.0	50.5	66.6	80.3	83.1	88.1	84.3	72.1	58.4	50.8	47.3	(19)
(20)		0.4%	MCWB	40.1	38.4	41.7	51.6	62.6	65.1	68.4	68.0	62.7	53.4	49.3	45.4	(20)
(21)		2%	DB	39.7	40.0	45.9	60.0	75.1	78.3	83.4	77.5	67.7	55.6	49.0	44.4	(21)
(22)		2%	MCWB	38.6	37.6	39.6	47.9	59.1	62.9	66.9	65.8	60.0	52.7	47.7	42.4	(22)
(23)		5%	DB	37.7	38.3	42.5	55.8	71.3	74.0	79.6	74.0	65.1	53.9	47.3	42.4	(23)
(24)		5%	MCWB	36.5	36.7	37.9	45.7	57.8	60.3	66.3	63.5	58.5	51.4	45.9	40.8	(24)
(25)		10%	DB	35.9	36.6	39.9	51.6	66.9	70.4	76.2	71.4	63.0	52.4	45.8	41.0	(25)
(26)		10%	MCWB	34.8	35.6	36.6	43.2	55.4	58.5	65.0	62.1	57.8	50.1	44.4	39.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.9	39.6	43.1	52.5	64.4	68.1	72.3	70.7	63.0	54.9	50.0	46.0	(27)
(28)		0.4%	MCDWB	41.0	41.1	47.9	64.9	74.2	79.9	81.5	77.8	68.4	56.4	50.4	46.9	(28)
(29)		2%	WB	38.8	38.2	40.4	49.1	61.8	64.6	70.1	67.7	61.5	53.6	47.8	43.0	(29)
(30)		2%	MCDWB	39.5	39.5	43.9	58.2	72.4	74.9	79.0	74.7	66.0	55.0	48.6	43.9	(30)
(31)		5%	WB	36.9	36.9	38.7	46.4	59.0	62.1	68.3	65.4	60.0	52.2	46.3	41.2	(31)
(32)		5%	MCDWB	37.6	38.1	41.5	54.0	69.9	71.5	76.6	71.5	64.0	53.6	47.2	42.2	(32)
(33)		10%	WB	35.1	35.7	37.1	44.2	55.9	60.0	66.2	63.8	58.6	50.5	44.7	39.8	(33)
(34)		10%	MCDWB	35.9	36.8	39.4	50.3	66.0	68.2	74.3	69.5	61.8	52.1	45.6	40.8	(34)
(35)	Mean Daily Temperature Range		MDBR	9.0	9.1	11.9	15.7	17.7	15.5	15.7	14.5	12.9	10.4	7.0	7.9	(35)
(36)		5% DB	MCDBR	5.9	8.9	14.1	23.9	24.4	22.1	20.8	18.3	15.8	10.8	7.3	8.4	(36)
(37)			MCWBR	5.7	7.4	9.6	13.6	11.7	10.0	8.2	8.4	9.1	8.4	6.8	7.7	(37)
(38)			MCDBR	6.0	8.2	12.5	20.8	22.8	19.1	17.4	14.9	13.2	9.1	7.1	7.8	(38)
(39)		5% WB	MCWBR	6.0	7.2	9.2	12.4	11.3	9.7	7.7	7.7	8.4	7.9	6.8	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.314	0.334	0.373	0.369	0.361	0.389	0.379	0.350	0.333	0.302	0.384	(40)	
(41)		taud	2.151	2.234	2.309	2.317	2.400	2.451	2.394	2.416	2.484	2.395	2.210	2.602	(41)	
(42)		Ebn at Noon	148	212	250	261	272	276	264	257	246	206	137	102	(42)	
(43)		Edh at Noon	13	22	28	33	33	32	33	30	24	19	12	9	(43)	
(44)	All-Sky Solar Radiation	RadAvg	75	261	741	1241	1693	1804	1690	1324	811	351	97	45	(44)	
(45)		RadStd	13	50	88	121	158	138	164	132	99	56	18	5	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	-239	N/A	N/A	N/A

Nomenclature: See separate page